[5] D. D. Streeter, "Gross morphology and fiber geometry of the heart," in *Handbook of Physiology*, R. M. Berne, Ed. Baltimore, MD: Williams & Williams, 1979, vol. 1, ch. 2: The Cardiovascular System, pp. 61–112.

[6] P. A. Helm, H.-J. Tseng, L. Younes, E. R. McVeigh, and R. L. Winslow, "*Ex vivo* 3D diffusion tensor imaging and quantification of cardiac laminar structure," *Magn. Reson. Med.*, vol. 54, pp. 850–859, 2005.

[7] R. Plonsey and R. C. Barr, "The four-electrode resistivity technique as applied to cardiac muscle," *IEEE Trans. Biomed. Eng.*, vol. BME-29, pp. 541–546, 1982.

[8] A. van Oosterom, R. W. de Boer, and R. T. Van Dam, "Intramural resistivity of cardiac tissue," *Med. Biol. Eng. Comput.*, vol. 17, pp. 337–343, 1979.

[9] P. Steendijk, G. Mur, E. T. van der Velde, and J. Baan, "The four-electrode resistivity technique in anisotropic media: Theoretical analysis and application on myocardial tissue *in vivo*," *IEEE Trans. Biomed. Eng.*, vol. 40, no. 11, pp. 1138–1147, Nov. 1993.

[10] P. Le Guyader, P. Savard, and F. Trelles, "Measurement of myocardial conductivities with an eight-electrode technique in the frequency domain," in *Proc. 17th IEEE-EMBS*, 1995, pp. 71–72.

[11] P. Le Guyader, F. Trelles, and P. Savard, "Extracellular measurement of anisotropic bidomain myocardial conductivities. I. Theoretical analysis," *Ann. Biomed. Eng.*, vol. 29, pp. 862–877, 2001.

[12] R. C. Barr and R. Plonsey, "Electrode systems for measuring cardiac impedances using optical transmembrane potential sensors and interstitial electrodes—Theoretical design," *IEEE Trans. Biomed. Eng.*, vol. 50, no. 8, pp. 925–934, Aug. 2003.

[13] L. Tung, "A bi-domain model for describing ischaemic myocardial DC potentials," Ph.D. dissertation, Massachusetts Inst. Technol., Cambridge, 1978.

[14] B. M. Johnston, P. R. Johnston, and D. Kilpatrick, "A new approach to the determinination of cardiac potential distributions: Application to the analysis of electrode configurations," *Math. Biosci.*, vol. 202, no. 2, pp. 288–309, 2006.

[15] R. M. Gulrajani, *Bioelectricity and Biomagnetism*. New York: Wiley, 1998.

[16] H. P. William, P. F. Brian, A. T. Saul, and T. V. William, *Numerical Recipes, The Art of Scientific Computing*, 2nd ed. Cambridge, MA: Cambridge Univ. Press, 1992.

[17] Y. Wang, P. H. Schimpf, D. R. Haynor, and Y. Kim, "Geometric effects on resistivity measurements with four- electrode probes in isotropic and anisotropic tissues," *IEEE Trans. Biomed. Eng.*, vol. 45, no. 7, pp. 8–19, Jul. 1998.

[18] A. E. Pollard and R. C. Barr, "Cardiac micro-impedance measurement in two-dimensional models using multisite interstitial stimulation," *Am. J. Physiol.-Heart Circulatory Physiol.*, vol. 290, no. 5, pp. H1976–H1987, 2006.

[19] F. X. Witkowski, K. M. Kavanagh, P. A. Penkoske, and R. Plonsey, "*In vivo* estimation of cardiac transmembane current," *Circ. Res.*, vol. 72, pp. 424–439, 1993.

[20] C.-S. Kim, S. Ufer, C. M. Seagle, C. L. Engle, H. T. Nagle, T. A. Johnson, and W. E. Cascio, "Use of micromachined probes for the recording of cardiac electrograms in isolated heart tissues," *Biosensors Bioelectron.*, vol. 19, no. 9, pp. 1109–1116, Apr. 2004.

[21] L. Clerc, "Directional differences of impulse spread in trabecular muscle from mammalian heart," *J. Physiol.*, vol. 255, pp. 335–346, 1976.

[22] D. E. Roberts, L. T. Hersh, and A. M. Scher, "Influence of cardiac fiber orientation on wavefront voltage, conduction velocity and tissue resistivity in the dog," *Circ. Res.*, vol. 44, pp. 701–712, 1979.

[23] D. E. Roberts and A. M. Scher, "Effects of tissue anisotropy on extracellular potential fields in canine myocardium *in situ*," *Circ. Res.*, vol. 50, pp. 342–351, 1982.

Study of On-Line Adaptive Discriminant Analysis for EEG-Based Brain Computer Interfaces

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***Abstract*—A study of different on-line adaptive classifiers, using various feature types is presented. Motor imagery brain computer interface (BCI) experiments were carried out with 18 naive able-bodied subjects. Experiments were done with three two-class, cue-based, electroencephalogram (EEG)-based systems. Two continuously adaptive classifiers were tested: adaptive quadratic and linear discriminant analysis. Three feature types were analyzed, adaptive autoregressive parameters, logarithmic band power estimates and the concatenation of both. Results show that all systems are stable and that the concatenation of features with continuously adaptive linear discriminant analysis classifier is the best choice of all. Also, a comparison of the latter with a discontinuously updated linear discriminant analysis, carried out in on-line experiments with six subjects, showed that on-line adaptation performed significantly better than a discontinuous update. Finally a static subject-specific baseline was also provided and used to compare performance measurements of both types of adaptation.**

***Index Terms*—AAR, automatic adaptive classification, band power estimates, BCI, Kalman filtering, LDA, on-line adaptation, QDA.**

I. Introduction

An electroencephalogram (EEG)-based brain computer interface (BCI) is a system which enables people to control devices using signals recorded from the scalp. One of the goals of such a device is to assist patients who have highly compromised motor functions, such as completely paralyzed patients with e.g., amyotrophic lateral sclerosis [1]–[11].

A BCI is divided in different modules: preprocessing, feature extraction, classification and feedback. Various signals are used in BCI systems, but our experiences were based in EEG signals, which can vary in time. Therefore, adaptation of modules like feature extraction or/and classification is a very important issue in BCI research. Regarding the classification module, up to now classifiers were manual and discontinuously updated after a number of runs, and the moment when to perform this upgrade was decided by the researcher depending on subjective factors such as his/her experience, [4], [12], [13]. Because of the disadvantage of this manual adaptation, recently several BCI research groups have started to work on automatic adaptation of the classifiers [14]–[18].

Our group presents in this paper results from on-line experiments with two different automatic adaptive classifiers and three different feature extraction techniques. The adaptation of the classifier is done in a

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Color versions of Figs. 4–6 are available online at http://ieeexplore.ieee.org.

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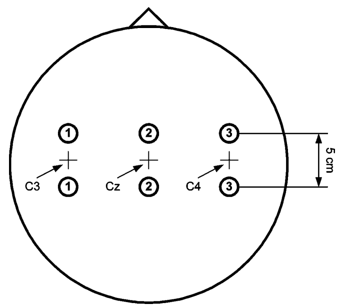

Fig. 1. Electrode Positions.

synchronous cue-based paradigm which allows supervised adaptation because the class is known "a priori". Supervised adaptive methods are mainly used for training inexperienced subjects who, in general, produce changing patterns. These changes can be due to their lack of experience and/or the use of different strategies. Supervised methods can also be used when a degradation in performance is observed in trained subjects, allowing to use the systems presented in this paper.

The first adaptive classifier we developed is based on the quadratic discriminant analysis (QDA) and the second on the linear one (LDA). Experiments were carried out with 18 naive able-bodied subjects. Each system was on-line tested with six subjects and simulated with the rest. Our goal was to find out which of the suggested systems had better performance. Therefore, we studied different features types: logarithmic band power (BP) estimates, adaptive autoregressive (AAR) parameters, and the concatenation of both (BPA).

We also performed experiments with six people using the concatenation of AAR and BP and a LDA classifier discontinuously updated after a number of runs, following the traditional scheme of a BCI. The purpose of these sessions was to ascertain whether continuously adaptive classifiers could perform better than discontinuously adaptive ones. In this case no simulation was performed because the discontinuous adaptation depends on subjective factors, such as the feeling of the subject to the feedback received with a specific classifier. A subject-specific static baseline was also provided and compared to the on-line adaptive and the discontinuously upgraded BCIs.

In summary, the aims of the paper are: to investigate which feature extraction method and which adaptive classifier is best for the framework presented; to find out if continuously adaptive classifiers can yield higher performance than discontinuously adaptive classifiers; to compare both with a calibrated subject-specific static baseline.

II. Methods

A. Feature Extraction

Three different feature types were used in the experiments: AAR parameters, logarithmic BP estimates and the concatenation of both in one vector. Six subjects received feedback with each of them.

1) AAR Parameters: Three parameters were calculated from each bipolar channel C3 and C4 (see Fig. 1), resulting a vector of six elements [20]–[24]. The parameters were calculated using Kalman filtering and the sliding window was 1/8 of 1 s.

2) Logarithmic BP Estimation: The signals of channels C3 and C4 were digitally bandpass filtered in the frequency bands 10–12 Hz and 16–24 Hz. Then they were squared, averaged over 1 s (sliding window), and a natural logarithm applied, [25]. The resulting vector had four elements.

3) Feature Concatenation: Both features types were then combined in one unique vector with ten elements, BPA.

B. Adaptive Classifiers

Two different classifiers were tested: adaptive information matrix (ADIM), which is an adaptive version of the QDA; Kalman adaptive LDA (KALDA), an adaptive version of the LDA based on Kalman filtering.

1) ADIM: QDA can be computed using the Mahalanobis distance of the feature vector $\mathbf{x}$ to each class i, $d_i(\mathbf{x})$, for which we need to know the mean value $\boldsymbol{\mu}_i$ and covariance matrix $\boldsymbol{\Sigma}_i$ of each class, as shown in (1)

$$d_i(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu}_i)^T \cdot \boldsymbol{\Sigma}_i^{-1} \cdot (\mathbf{x} - \boldsymbol{\mu}_i) \quad (1)$$

$$D(\mathbf{x}) = \sqrt{d_1(\mathbf{x})} - \sqrt{d_2(\mathbf{x})}. \quad (2)$$

For classifying the feature vector one looks to the sign of $D(\mathbf{x})$, assigning it to class 2 if it is greater than zero and to class 1 otherwise. The on-line updating of the classifier can be done by estimation of the inverse of the covariance matrix of each class i with the "Matrix Inversion Lemma" [27]

$$\boldsymbol{\Sigma}_{k,i}^{-1} = (1 + UC) \cdot \boldsymbol{\Sigma}_{k-1,i}^{-1} - \frac{1 + UC}{1 - UC + \mathbf{x}_{k,i} \cdot \mathbf{v}} \cdot \mathbf{v} \cdot \mathbf{v}^T \quad (3)$$

where $\mathbf{v} = \boldsymbol{\Sigma}_{k-1,i}^{-1} \cdot \mathbf{x}_{k,i}$, UC is the update coefficient for the adaptation, $\mathbf{x}_{i,k}$ is the current class i feature vector, k is the current sample and i is the class label. The mean vector $\boldsymbol{\mu}_i$ is also needed for the computation of $D(\mathbf{x})$ and needs to be estimated. This mean vector was incorporated as additional row and column to the matrix $\boldsymbol{\Sigma}_i$ for its automatic estimation and to avoid an extra algorithm, resulting in an "extended" covariance matrix. For more information see [32].

The starting matrix $\boldsymbol{\Sigma}_0$ was computed from 1620 trials recorded from seven subjects in different sessions. The data were recorded using different paradigms, [12].

2) KALDA: The equation for the LDA classification of data of two different classes can be written as

$$D(\mathbf{x}) = b + \mathbf{w}^T \cdot \mathbf{x} \quad (4)$$

$$b = -\mathbf{w}^T \cdot \frac{1}{2} \cdot (\boldsymbol{\mu}_2 + \boldsymbol{\mu}_1) \quad (5)$$

$$\mathbf{w} = (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1} \cdot (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1) \quad (6)$$

where $\boldsymbol{\Sigma}_i$ is the covariance matrix for class i, $\boldsymbol{\mu}_i$ is the mean value for class i, and $\mathbf{x}$ is the feature vector. Similarly to QDA, if $D(\mathbf{x})$ is greater than zero, then the input is classified as class 2, and if the output is equal or less than zero, then the input is classified as class 1.

LDA is a weight vector $[b, \mathbf{w}^T]$ which can be estimated using Kalman filtering, in which the Kalman gain [see (11) and (12)] varies the update coefficient and changes the adaptation speed depending on the properties of the data. In comparison to ADIM [see $\boldsymbol{A}_k$ in Kalman (7)–(14)], only one matrix needs to be estimated and all trials can be used to update the weight vector. The update equations for Kalman filtering can be summarized as follows:

$$e_k = y_k - \mathbf{H}_k \cdot \widehat{\boldsymbol{w}}_{k-1} \quad (7)$$

$$\mathbf{H}_k = \left[1, \mathbf{x}_k^T\right] \quad (8)$$

$$Q_k = \mathbf{H}_k \cdot \mathbf{A}_{k-1} \cdot \mathbf{H}_k^T + v_k \quad (9)$$

$$\mathbf{k}_k = \frac{\mathbf{A}_{k-1} \cdot \mathbf{H}_k^T}{Q_k} \quad (10)$$

$$\widehat{\boldsymbol{w}}_k = \widehat{\boldsymbol{w}}_{k-1} + \mathbf{k}_k \cdot e_k \quad (11)$$

$$\tilde{\boldsymbol{A}}_k = \mathbf{A}_{k-1} - \mathbf{k}_k \cdot \mathbf{H}_k \cdot \mathbf{A}_{k-1} \quad (12)$$

$$\mathbf{A}_k = \frac{\operatorname{trace}(\tilde{\boldsymbol{A}}_k) \cdot UC}{p} + \tilde{\boldsymbol{A}}_k \quad (13)$$

$$v_k = 1 - UC. \quad (14)$$

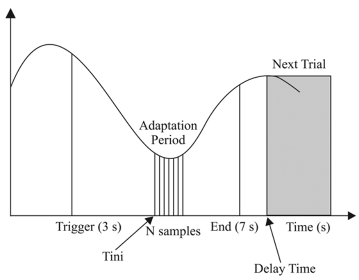

Fig. 2. Adaptation scheme within a trial.

where e_k is the one-step prediction error, y_k is the current class label, $\mathbf{H}_k$ is the measurement matrix, $\widehat{\boldsymbol{w}}_k$ is the state vector (the estimated weights for the LDA), $\mathbf{x}_k$ is the current feature vector, Q_k is the estimated prediction variance, $\tilde{\boldsymbol{A}}_k$ is an intermediate value needed to compute $\mathbf{A}_k$ which is the a priori state error correlation matrix, v_k is the variance of the innovation process, $\mathbf{k}_k$ is the Kalman gain, UC is the update coefficient and p is the number of elements of $\widehat{\boldsymbol{w}}_k$.

The starting values $\boldsymbol{A}_0$ and $\boldsymbol{w}_0$ were computed from the same 1620 trials used to estimate the initial conditions of ADIM.

C. Other Parameters

Still other elements have to be defined to describe the adaptation. These parameters are defined for synchronous experimental paradigms, as the one described in this paper (see Section II-F).

- Initial Time, *Tini* of the adaptation in each trial. It can be defined by selecting the time of the maximum mutual information (MI) [28] of the last three runs (see Section II-F). MI is a measurement that takes into account not only the sign of the classifier output, but also its magnitude. It is based on the principle of Shannon's communication theory, and it is defined as the average amount of information that the classification output provides about the signal, given that the classification output can be considered as consisting of signal and noise. Another method to estimate *Tini* is to continuously update it by an on-line estimation of the MI, $\widehat{\mathbf{MI}}_t$, using a moving average algorithm, shown in (15)

$$\widehat{\mathbf{MI}}_t = \mathbf{mi} \cdot UC_{tini} + \widehat{\mathbf{MI}}_{t-1} \cdot (1 - UC_{tini}) \quad (15)$$

$$Tini = t|_{\max(\widehat{\mathbf{MI}}_t)} \quad (16)$$

  where mi is the output of the classifier multiplied with the class label of the current trial, and UC_{tini} is the update coefficient for *Tini* estimation. The time of maximum $\widehat{\mathbf{MI}}_t$ is selected as *Tini* for the next trial.
- Window length, N is the number of samples of the adaptation period, here the value was fixed to 125 samples (1 s).
- $DelayTime$ is defined because the updated classifier needs to start after the end of the current trial so that the samples used for the adaptation are independent from the ones when the classifier is applied.

The parameters UC, UC_{tini}, N were found by optimization on six subjects for which all sets of parameters were evaluated. More information about this optimization can be found in [26]. Fig. 2 is a scheme of the adaptation within a trial, and it is repeated in every trial of each run. As it is shown, the classifier is updated using a window of N starting at *Tini*, where *Tini* is estimated either in a discontinuous or in a continuous manner.

D. Systems With Continuously Adaptive Classifiers

Three systems were tested and distinguished by the feature and classifier type used. We used three kind of feature extraction methods: AAR, BP, and BPA. We decided to try BPA in order to avoid the "*a priori*" feature type selection of AAR or BP, as we did not have any information about the subject EEG before the experiments. Some subjects happen to have better results with one or the other feature type, but the use of BPA solved this problem. We also did experiments with continuously adaptive QDA and LDA. In this paper, the suggested adaptive LDA is a more developed version of our adaptive QDA because UC is not fixed. Due to the number of combinations of feature types and classifiers, the systems were simulated to reduce the number of subjects.

1) System 1: The features used were AAR parameters and the classifier was ADIM. For subjects S01 to S03 initial time was discontinuously updated every three runs (Sys 1. DTini). For subjects S04 to S06, it was continuously adapted (Sys 1. CTini). For simulation DTini was used with subjects S07 to S09 and CTini with subjects S10 to S18.

2) System 2: Logarithmic BP estimates features and ADIM classifier were selected. For subjects S07 to S09 the initial time was updated every three runs (Sys 2. DTini) and for subjects S10 to S12 it was continuously adapted (Sys 2. CTini). For simulation DTini was used with subjects S01 to S03 and CTini with subjects S04 to S06 and S13 to S18.

3) System 3: The selected features were the concatenation of BP estimates and AAR parameters and the classifier was KALDA. The initial time for adaptation was continuously estimated (CTini). This system was tested with subjects S13 to S18.

4) System 4: The concatenation of BP estimates and AAR parameters were classified by ADIM with continuously adaptive *Tini* (CTini). This system was only off-line simulated for the complete analysis of the results.

TABLE I
ON-LINE AND SIMULATED SYSTEMS WITH EACH SUBJECT

Subj	Sys.1	Sys.2	Sys.3	Sys.4
S01-S06	on-line	simulated	simulated	simulated
S07-S12	simulated	on-line	simulated	simulated
S13-S18	simulated	simulated	on-line	simulated

E. Simulation

The on-line and simulated systems for each subject are summarized in Table I.

The results of systems 1, 2, and 3 were obtained on-line (6 subjects) and by simulation (12 subjects), making the analysis balanced for them. As one can expect a bias towards the method used for feedback (one third of the total), significant differences were found in disadvantageous circumstances. On the other hand system 4 was only simulated for analysis of the feature type; therefore the comparison is disadvantageous if one tries to discover whether system 4 is significantly better than the rest of the systems, and we focused in this outcome.

F. Description of the Experiments

The experiments were carried out with 18 able-bodied subjects without previous BCI experience. A total of 24 subjects started the experiments, but only 18 of them finished. From the six subjects who did not finish the experiments, four did not have separable features in the standard frequency bands and two of them caused too many artefacts during the feedback period.

The recording was made using a g.tec amplifier (Guger Technologies OEG Austria) and Ag/AgCl electrodes. Three bipolar EEG channels were measured over the positions C3, Cz and C4 (see Fig. 1). The EEG was filtered between 0.5 and 30 Hz and sampled with 125 Hz.

The experiment consisted of three sessions for each subject. Each session was conducted on a different day and nine runs were recorded. One run consisted of 40 feedback trials. Between runs there was a break of some minutes (between 1 and 15 min). For each subject the total number of trials is 1080.

The experiment was based on the basket paradigm [29]. The subjects sat on a relaxing chair with armrests. In each trial the subject saw a black screen for a fixed length pause (3 s). Then two different colored baskets (green and red) appeared at the bottom of the screen. At this moment, also a little green ball appeared at the top of the screen. After 1 s more, the ball began to fall downward with constant speed. The horizontal position of the ball was directly controlled by the output of the continuous adaptive classifier, providing continuous visual feedback for 3 s. The subject's task was to control the green ball by the imagination of left or right hand movements, [30], [31], and try to keep it as long as possible in the side where the green basket appeared. The duration of each trial was 7 s, with a random interval between trials from 1 to 2 s. The order of left and right cues was random. The subjects were instructed to only imagine the movements and they were said that real movements were forbidden. A camera mounted in front of the subject was used to check that he/she kept still. Also during the experiment, the recorded signals were displayed in a screen to visually detect artifacts and to check if the subjects used them to control the system. However, in [31] it was published that EEG patterns of performed and imagined movements are similar. No automatic artifacts rejection was applied during the experiments. We instructed the subjects to keep the green ball in the side where the green basket appeared, as long as possible, in order to find at which moment of the trial the separation between classes was the best.

G. Subject-Specific Static Baseline Versus Discontinuous Classifier Versus Continuous Classifier Adaptation

An important issue that arises when studying the topic of adaptive classifiers is whether continuously adaptive classifiers are better than a discontinuously adaptive ones. Discontinuously adaptive classifiers have been always used in BCI research, and as explained in Section I, the decision when to update them was mainly based on the researcher's experience and the subject's control; therefore it is not easy to simulate such a system and on-line experiments are preferable. Taking this into account, we performed on-line experiments with a discontinuously adaptive classifier, using the basket paradigm as presented in Section II-F. The features were BPA and the classifier LDA. Six subjects took part in the experiments and they were all naive.

For these experiments the general scheme when to update the classifier was as follows: the first classifier used with every subject was the general classifier. After three runs, the LDA classifier was updated (using these three runs) and used for the next three runs. Then a classifier using the six runs already recorded was calculated for the next three runs. For the next session, the classifier of the session carried out the day before was calculated and used (9 runs). The day after again three, six, and nine runs were used for the calculation of the classifier and so on. However, this scheme could vary depending on the feedback received with the updated classifier. When the feedback was worse and/or the subject felt less control with the new classifier than with the old one, the run was stopped and started again with the classifier which seemed to provide better feedback. In any case, the minimum amount of runs for the calculation of the classifier was 3.

Besides, we provide a subject-specific static baseline for comparison of the two adaptive systems for which we used BPA features and LDA-based classifier. Six naive subjects performed experiments in three days. Each day 3 nonfeedback runs were recorded, then a subject-specific classifier was calculated and used to classify six feedback runs. We did not change the classifier during the feedback runs. The feedback performance of these subjects in each session served as baseline for comparison with the two adaptive systems.

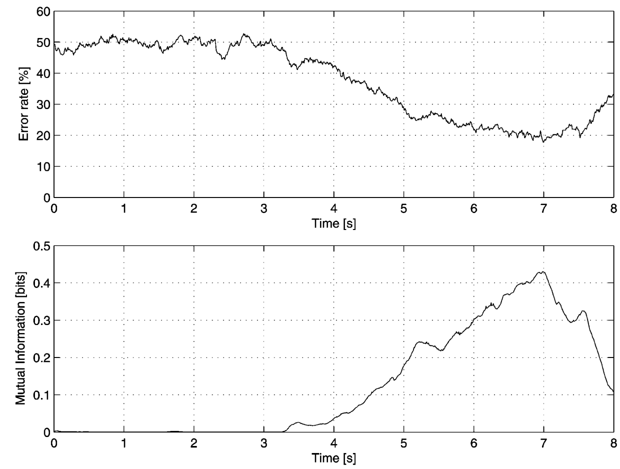

Fig. 3. Single trial analysis of Subject S03 in session 3. ERR and MI for ADIM are shown respectively above and below.

III. Results

A. Experimental Results

Fig. 3 displays an example of a single trial analysis of one subject in one experimental session. In Fig. 3, the Error Rate (ERR) [%] versus time [s] and MI [bit] versus time [s] for subject S03 in the third session are depicted. It is obtained computing a single trial analysis of the session. The subfigure above shows the mean ERR and the subfigure below is the mean MI over all trials of the session.

For each subject and session (360 trials) the single trial analysis, as shown in Fig. 3, was performed using the output of the corresponding classifier. Then, the minimum ERR and maximum MI were selected to describe its performance and to perform relative comparisons between the different methods presented in this paper; this is in contrast with methods used in real applications where the decision time is defined beforehand and the subject's performance is computed at a predefined time instant or interval. Fig. 4 shows the mean performance and standard error of the mean (SEM) reached with each system in each experimental session.

In Fig. 4 the system with best mean performance is number 3, followed by system 4. Up to the second session, system 1 performed better than system 2 in terms of ERR, but did not provide better MI.

It was interesting for us to see if the different ways of updating *Tini* had an impact in the performance of the system. In Fig. 5, the average performance and SEM of CTini and DTini are depicted. In order to use the same number of observations for each method, only results from systems 1 and 2 and subjects S01 to S12 were used; therefore six feedback and six simulated performance results were available in each session. As we see the mean performance was better for CTini than for DTini in every session.

B. Subject-Specific Static Baseline Versus Discontinuous Classifier Versus Continuous Classifier Adaptation

Fig. 6 depicts the "learning curve" of the subjects who used LDA, KALDA or did not receive any adaptation (static).

Fig. 6 shows that subjects who did not receive adaptation had much worse performance than the rest. Subjects who used an adaptive system showed improvement from session 1 to session 3, and this improvement was greater for the automatically adaptive system. The major difference between the manual and the automatic adaptive systems was found in session 3, for both ERR and MI.

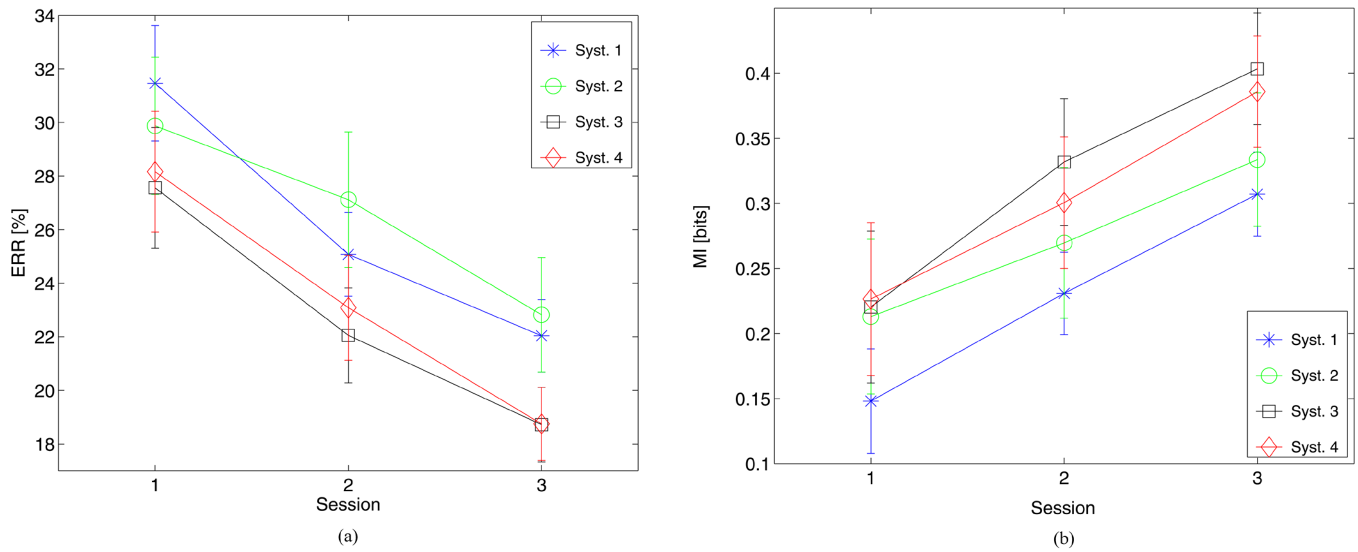

Fig. 4. Average performance reached with each system in each experimental session. (a) Average and SEM of minimum ERR. (b) Average and SEM of maximum MI.

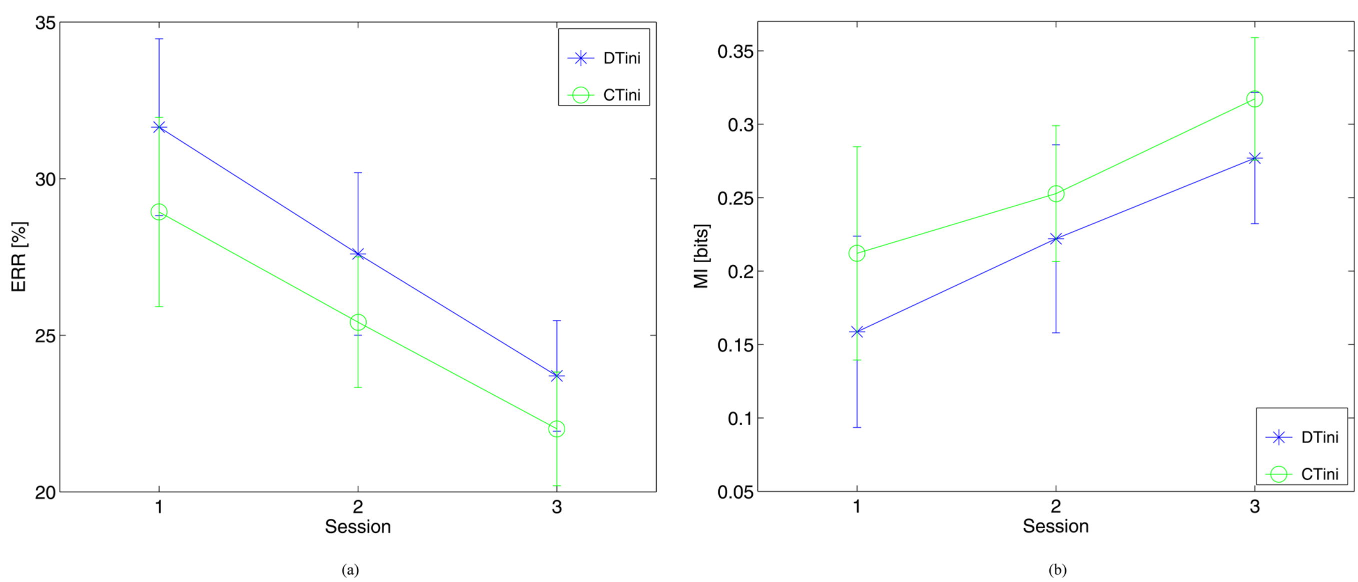

Fig. 5. Average performance reached using DTini or CTini. (a) Average and SEM of minimum ERR. (b) Average and SEM of maximum MI.

IV. Discussion

A. Experimental Results

In Fig. 4 it is shown that systems 1 and 2 perform very similar and their performance is worse than for systems 3 and 4. The use of AAR parameters (system 1) or logarithmic BP estimates (system 2), yielded similar performance in our experiments. On the other hand, system 2 performed better than one in terms of MI. This is probably because the sliding window used to calculate the AAR parameters was eight times smaller than the one used to estimate the BP. In Fig. 4 we can see that BP estimates have higher MI values than AAR parameters, even when the ERR is similar. Systems 3 (BPA+KALDA) and 4 (BPA+ADIM) show better performance (in terms of ERR and MI) than systems 1 and 2. Systems 3 and 4 use the same kind of features, BPA, with a different adaptive classifier. Therefore it can be interpreted that the improvement in performance from systems 1 and 2 to 3 and 4 is due to the feature concatenation.

In Fig. 4 it is also shown that system 1 performs worst in the first day. This is probably related to the initial classifier with AAR parameters, which had a minimum ERR of approximately 37%. On the other hand the initial classifier for system 2 had a minimum of approximately a 32% and a 33% for system 3. It can also be seen that system 2, with BP estimates is the one which improves less. The BP estimates are fixed in the mu and beta frequency bands. This could be a reason why the performance improvement is smaller. We know that the EEG is nonstationary as showed in [18], [19], [26] and therefore having fixed bands could be a disadvantage. In fact, as we can see in Fig. 4 when an adaptive element is added to the fixed frequency bands, the performance improves significantly.

Fig. 5 shows better mean values for CTini than for DTini in each experimental session and for both performance measurements, but the differences between them were not found significant. This means that both methods for *Tini* estimation are equivalent.

B. Subject-Specific Static Baseline Vs. Discontinuous Classifier Vs. Continuous Classifier Adaptation

Fig. 6 shows the difference between discontinuous, continuous and no adaptation over three sessions. The difference between the

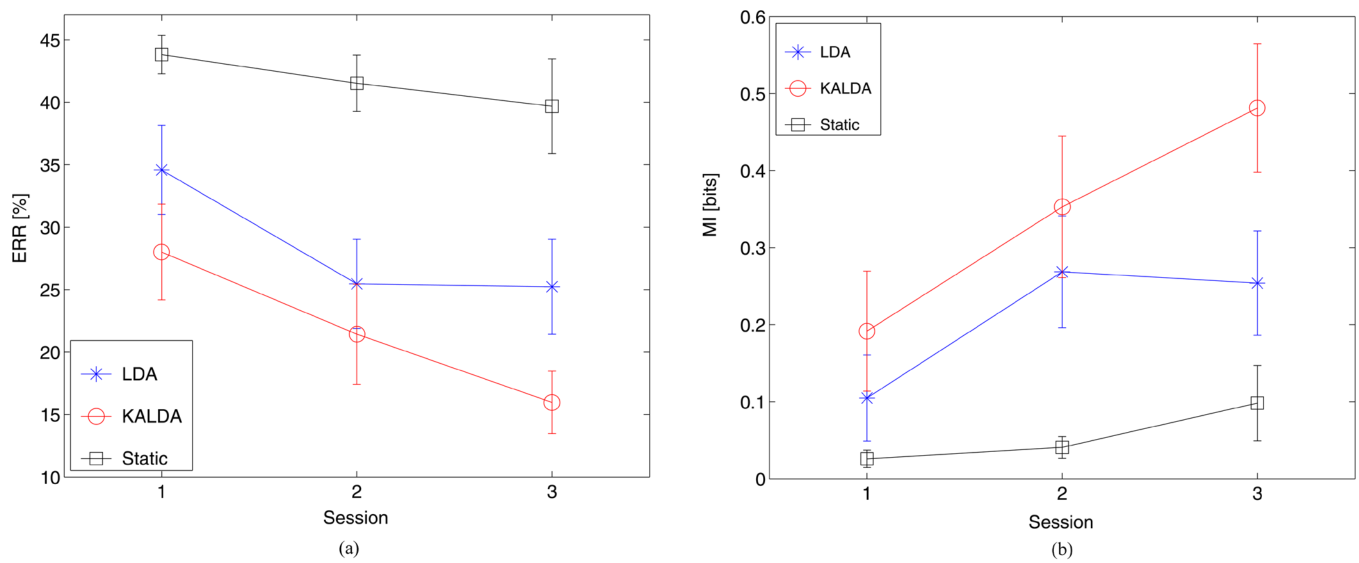

Fig. 6. Average performance reached using KALDA, manual LDA adaptation or no adaptation. (a) Average and SEM of minimum ERR. (b) Average and SEM of maximum MI.

static baseline of no adaptation and the manual updating was found significant in sessions 2 and 3 for the ERR and session 2 for the MI. The continuously adaptive system was found significantly better than the nonadaptive in every session for both performance measurements. The static baseline with daily tuning showed much worse performance than the systems which started with a general classifier and were then adapted. This is especially the case of the continuous adaptive system, which was significantly better than the static baseline in every session. Some reasons for the bad performance of the subjects of the static system can be that the nonfeedback data used to tune the classifier differ from the data recorded in feedback runs, [18] and that under this framework, the subject has to adapt to the classifier to have a good performance, which is a difficult task especially for inexperienced subjects.

No significant difference was found between the two adaptive systems in sessions 1 and 2, but the continuously adaptive systems was found significantly better than the discontinuously adaptive one in session 3 for both ERR and MI. The discontinuous adaptation showed no improvement between session 2 and 3. The reason for this outcome could be explained by the Man-Machine Learning Dilemma (MMLD), [30]. It should be expected that in BCI feedback experiments the classification accuracy improves with increasing number of sessions. However, as shown in this paper, this does not always happen. As described in [30] MMLD "implies that two systems (man and machine) are strongly interdependent but have to be adapted independently". With the adaptive continuous adaptation, in both features and classifier, the classification accuracy improved in every session and in the presence of feedback. The continuous adaptation might improve the harmful effect of the MMLD with the use of a stable trial-based update of the BCI system, as the two interdependent systems can be now dependently adapted.

V. Conclusion

In this paper, several elements have been described: feature concatenation, continuously adaptive classifiers and a comparison between them and the traditional discontinuously adaptive classification and a subject-specific static baseline. All adaptive methods presented are supervised (true labels are known) and were tested with inexperienced subjects who in general produce changing patterns. This type of methods could also be used for re-training the interface when a decrease in performance is observed. In real BCI applications the labels are unknown and the methods presented here should be modified, issue in which we are currently working.

Within the framework of the systems used in this paper (two bipolar channels, two-class systems), the results suggested three conclusions: feature concatenation of BP and AAR is the best choice of the ones presented, both KALDA and ADIM are stable continuously adaptive classifiers and perform similarly and, from experiments performed with 12 subjects, continuously adaptive classifiers showed statistically significant better performance than discontinuously adaptive ones. Therefore, under the settings presented in this paper, the use of continuously adaptive classifiers is justifiable and even advisable.

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References

[1] G. Pfurtscheller, D. Flotzinger, and J. Kalcher, "Brain-computer interface —A new communication device for handicapped people," *J. Microcomput. Appl.*, pp. 293–299, 1993.

[2] N. Birbaumer, N. Ghanayim, T. Hinterberger, I. Iversen, B. Kotchoubey, A. Kübler, J. Perelmouter, E. Taub, and H. Flor, "A brain-controlled spelling device for the completely paralyzed," *Nature*, vol. 398, pp. 297–298, 1999.

[3] T. M. Vaughan, J. R. Wolpaw, and E. Dochin, "EEG-based communication: Prospects and problems," *IEEE Trans. Rehabil. Eng.*, vol. 4, no. 4, pp. 425–430, Dec. 1996.

[4] B. Blankertz, G. Curio, and K. R. Müller, "Classifying single trial EEG: Towards brain computer interfacing," in *Advances in Neural Information Processing Systems 14*, T. G. Diettrich, S. Becker, and Z. Ghahramani, Eds. Boston, MA: MIT-Press, 2002, pp. 157–164.

[5] J. R. Wolpaw and D. J. McFarland, "Multichannel EEG-based brain-computer communication," *Electroenceph. Clin. Neurophysiol.*, vol. 90, pp. 444–449, 1994.

[6] J. R. Wolpaw, N. Birbaumer, W. J. Heetderks, D. J. McFarland, P. H. Pekham, G. Schalk, E. Dochin, L. A. Quatrano, C. J. Robinson, and T. M. Vaughan, "Brain-computer interface technology: A review of the first international meeting," *IEEE Trans. Rehabil. Eng.*, vol. 8, no. 2, pp. 164–173, Jun. 2000.

[7] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, "Brain-computer interfaces for communication and control," *Clin. Neurophysiol.*, vol. 113, pp. 767–791, 2002.

[8] G. Pfurtscheller, C. Neuper, G. Müller, B. Obermaier, G. Krausz, A. Schlögl, R. Scherer, B. Graimann, C. Keinrath, D. Skliris, M. Woertz, G. Supp, and C. Schrank, "Graz BCI: State of the art and clinical applications," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 11, no. 2, pp. 177–180, Jun. 2003.
[9] K. R. Müller, C. W. Anderson, and G. E. Birch, "Linear and nonlinear methods for brain-computer interfaces," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 11, no. 2, pp. 165–169, Jun. 2003.
[10] N. Birbaumer, T. Hinterberger, A. Kübler, and N. Neumann, "The thought-translation device (TTD): Neurobehavioral mechanisms and clinical outcome," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 11, no. 2, pp. 120–123, Jun. 2003.
[11] J. del R. Millán, "Brain-computer interfaces," in *The Handbook of Brain Theory and Neural Networks*, M. A. Arbib, Ed., 2nd ed. Cambridge, MA: MIT Press, 2002, pp. 178–181.
[12] G. Pfurtscheller, C. Neuper, C. Guger, W. Harkam, H. Ramoser, A. Schlögl, B. Obermaier, and M. Pregenzer, "Current trends in graz brain-computer interface (BCI) research," *IEEE Trans. Rehabil. Eng.*, vol. 8, no. 2, pp. 216–219, Jun. 2000.
[13] D. J. McFarland, A. T. Lefkowicz, and J. R. Wolpaw, "Design and operation of an EEG-based brain-computer interface with digital signal processing technology," *Behavior Res. Meth., Instrum., Comput.*, vol. 29, pp. 337–345, 1997.
[14] J. del R. Millán, "On the need for on-line learning in brain-computer interfaces," presented at the Int. Joint Conf. Neural Networks, Budapest, Hungary, 2004.
[15] T. Nykopp, J. Heikkonen, and M. Sams, "Sequential classification of finger movements from MEG recordings," in *Biomedisinische Technik, Proc. 2nd Int. Brain-Computer Interface Workshop and Training Course (Graz)*, 2004, pp. 77–78.
[16] A. Buttfield, P. W. Ferrez, and J. del R. Millán, "TTowards a robust BCI: error potentials and online learning," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 14, no. 2, pp. 164–168, Jun. 2006.
[17] P. Sykacek, S. J. Roberts, and M. Stokes, "Adaptive BCI based on variational bayesian kalman filtering: An empirical evaluation," *IEEE. Trans. Biomed. Eng.*, vol. 51, no. 5, pp. 719–727, May 2004.
[18] M. Kraudelat, P. Shenoy, B. Blankertz, R. P. N. Rao, and K.-R. Müller, "Online adaptation for CSP based BCI systems," presented at the Brain-Computer Interface Technology: 3rd Int. Meeting, Rensselaerville, NY, Jun. 14–19, 2005.
[19] A. Schlögl, C. Vidaurre, and G. Pfurtscheller, "Assessing non-stationarities in BCI data," presented at the Brain-Computer Interface Technology: 3rd Int. Meeting, Rensselaerville, NY, Jun. 14–19, 2005.
[20] C. Neuper, A. Schlögl, and G. Pfurtscheller, "Enhancement of left-right sensorimotor EEG differences during feedback-regulated motor imagery," *J. Clin. Neurophysiol.*, vol. 16, pp. 373–382, 2000.
[21] A. Schlögl, D. Flotzinger, and G. Pfurtscheller, "Adaptive autoregressive modeling used for single-trial EEG classification," *Biomed. Technik*, vol. 42, pp. 162–167, 1997.
[22] A. Schlögl, *The Electroencephalogram and the Adaptive Autoregressive Model: Theory and Applications*. Aachen, Germany: Shaker Verlag, 2000.
[23] G. Pfurtscheller, C. Neuper, A. Schlögl, and K. Lugger, "Separability of EEG signals recorded during right and left motor imagery using adaptive autoregressive parameters," *IEEE Trans. Rehabil. Eng.*, vol. 6, no. 3, pp. 316–325, Sep. 1998.
[24] C. Guger, A. Schlögl, C. Neuper, C. Walterspacher, D. Strein, T. Pfurtscheller, and G. Pfurtscheller, "Rapid prototyping of an EEG-based brain-computer interface (BCI)," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 9, no. 1, pp. 49—58, Mar. 2001.
[25] G. Pfurtscheller, C. Neuper, D. Flotzinger, and M. Pregenzer, "EEG-based discrimination between imagination of right and left hand movement," *Electroenceph. Clin. Neurophysiol.*, vol. 103, pp. 642–651, 1997.
[26] C. Vidaurre, A. Schlögl, R. Cabeza, R. Scherer, and G. Pfurtscheller, "A fully on-line adaptive BCI," *IEEE Trans. Biomed. Eng.*, vol. 53, no. 6, pp. 1214–1219, Jun. 2006.
[27] S. Haykin, *Adaptive Filter Theory*. Upper Saddle River, NJ: Prentice Hall, 1996.
[28] A. Schlögl, C. Neuper, and G. Pfurtscheller, "Estimating the mutual information of an EEG-based brain-computer interface," *Biomed. Technik*, vol. 47, pp. 3–8, 2002.
[29] G. Krausz, R. Scherer, G. Korisek, and G. Pfurtscheller, "Critical decision-speed and information transfer in the graz brain-computer-interface," *Appl. Psychophysiol. Biofeedback*, vol. 28, pp. 233–240, Sep. 2003.
[30] G. Pfurtscheller and C. Neuper, "Motor imagery and direct brain-computer communications," *Proc. IEEE*, vol. 89, no. 7, pp. 1123–1134, Jul. 2001.
[31] ——, "Motor imagery activates primary sensorimotor area in humans," *Neurosci. Lett.*, vol. 239, pp. 65–68, 1997.
[32] C. Vidaurre, "On-line adaptive classification for brain-computer interfaces," Ph.D. dissertation, Public Univ. Navarre, Navarre, Spain, 2006.

Estimation of the Uncertainty in Time Domain Indices of RR Time Series

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***Abstract*—A method for estimating the uncertainty in time-domain indices of RR time series is described. The method relies on the central limit theorem that states that the distribution of a sample average of independent samples has an uncertainty that asymptotically approaches to the sample standard deviation divided by the square root of the number of samples. Because RR time series cannot be characterized by a set of independent samples, we propose to estimate the uncertainty of indices by computing them in blocks that satisfy that the obtained partial indices are independent. We propose a methodology to search sets of independent partial indices and apply this methodology to the estimation of the uncertainty in the mean RR, SDRR, and r-msDD indices. The results show that the uncertainty can be higher than the 10% of the index for the SDRR and even higher for the r-msDD. Moreover, a statistical test for the difference of two indices is proposed.**

I. Introduction

Heart rate variability (HRV) is regarded as a marker of the relationship between the cardiovascular and autonomic nervous systems (see the Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology [1], for a good review on HRV applications and recommendations for measurements). There are several techniques that quantify the beat-to-beat fluctuations although, nowadays, the time-domain analysis is the most employed methodology in clinical studies, both in short-term and 24-h recordings.

Besides of applications of time domain analysis of HRV based on 24-h recordings, such as prediction of cardiac disease and mortality [2]–[4], HRV analysis from short-term recordings is becoming a powerful tool for the prediction of several diseases (mortality prediction after myocardial infarction [5], studies on diabetic patients [6], assessment of training status in athletes [7], or assessment of psychological stress [8]). Among the whole proposed time-domain indices, the most recurrently calculated are the mean $(\mathrm{RR}_{\mathrm{mean}})$, the standard deviation

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